



Weekly Progress Report

Nanoelectronics Integrated Systems Center (NISC)
Research Internship

Week 1 Report

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Research Topic: Benchmarking LLMs for Exploratory Data Analysis
on Chaotic Time-Series Datasets
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Week: Week 1

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Overview

The first week of the internship focused on building a strong theoretical foundation for the research Area. During the first days, we studied the fundamentals of Chaos Theory, scientific research methodology, nonlinear analysis metrics, and the current state of Large Language Models (LLMs) for Exploratory Data Analysis (EDA). In addition, an initial literature review was initiated to understand existing benchmark frameworks and identify potential research directions.

Day 1: Research Fundamentals and Chaos Theory

The first day was dedicated to understanding the fundamentals of scientific research and the basic principles of Chaos Theory.

The following topics were studied:

- Fundamentals of scientific research.
- Characteristics of high-quality research papers.
- Impact Factor (IF), H-index, and citation metrics.
- Scientific publishers and digital libraries including IEEE, Springer, Elsevier, ACM Digital Library, Nature, and arXiv.
- Effective strategies for searching and filtering research papers.
- Introduction to Chaos Theory and nonlinear dynamical systems.
- Applications of chaotic systems in engineering and artificial intelligence.

Day 2: Chaos Metrics and Practical Implementation

The second day focused on understanding quantitative measures used to characterize chaotic systems.

Several nonlinear metrics were studied, including:

- Lyapunov Exponents.
- Correlation Dimension.
- Entropy-based measures.
- Other nonlinear complexity metrics.

we used python to implement and experiment with several of these metrics. Practical exercises were conducted to understand how these measurements are computed and how they can be used to analyze nonlinear and chaotic time-series data.

Day 3: Starting Search about LLMs for Data Analysis

The third day was devoted to reviewing the recent literature related to Large Language Models in data analysis tasks.

The following activities were completed:

- Searching for research papers related to LLMs, EDA, and data science assistants.
- Creating the first version of a literature review spreadsheet containing:
 - Paper title
 - Authors
 - Publication venue
 - Publication year
 - Publisher
- Reading and summarizing multiple papers related to:
 - LLM Benchmarking
 - Exploratory Data Analysis
 - Automated Data Analysis
 - Tabular Reasoning
 - Data Science Agents
- Identifying several benchmark frameworks designed for evaluating LLMs on data analysis and data science tasks.

This literature review serves as the foundation for identifying appropriate evaluation methodologies and benchmark datasets for the proposed research.

Day 4: EDA on Time-Series and Chaotic Datasets

The fourth day focused on understanding how Exploratory Data Analysis is performed on time-series datasets and investigating how this process differs when dealing with chaotic datasets.

Compared the results of Claude Sonnet Medium on the SCADA Wind Speed dataset using two different system prompts: one explicitly indicating that the dataset is chaotic and another without mentioning its chaotic nature

Progress Summary

The first four days established the theoretical and practical foundation required for the research project. Significant progress was achieved in understanding Chaos Theory, nonlinear analysis metrics, scientific research methodologies, and the current landscape of LLM-based data analysis.